Project 3: Cross-Lingual Sentiment and Intent Alignment for Low-Resource Languages

Problem Description

Develop a robust natural language understanding system to accurately detect sentiment and underlying intent in regional or morphologically rich Indian language text.

The system must bridge the gap between high-resource languages and low-resource languages, aligning semantic meaning without relying on direct word-for-word translation.

Input and Output Specifications

- **Input:** Raw, unstructured text samples in one or more regional languages, sourced from domains such as news, social media, or conversational dialogue.
- **Output:** A predicted classification label for sentiment, such as positive, negative, or neutral, and a separate predicted label for intent, such as inquiry, opinion, or factual report.
- **Confidence Metrics:** The system must also output probability/confidence scores for each predicted category.

Technical Guidelines and Freedom

3.3.1 Data Autonomy

Students have complete freedom to curate, scrape, or select any publicly available multilingual or cross-lingual dataset.

3.3.2 Architecture Choice

The choice of the underlying model is entirely open. Any transformer-based architecture can be trained from scratch or fine-tuned.

3.3.3 Resource Efficiency

The entire data preprocessing, training, and inference pipeline must be optimized to run within the memory constraints of a standard free-tier GPU such as Kaggle or Google Colab.

Expected Deliverables

- A modular, well-documented notebook containing the end-to-end training and inference pipeline.
- An evaluation report demonstrating the model's performance on the target languages using standard metrics such as F1-score and accuracy.
- A brief analytical summary discussing the model's cross-lingual transfer capabilities and how it handled specific linguistic complexities.

Dataset

Students can select any suitable publicly available multilingual or cross-lingual dataset containing sentiment and/or intent annotations for the selected languages.

`https://aikosh.indiaai.gov.in/home/datasets/details/samanantar_largest_parallel_corpus_for_indic_languages.html`